

**Writing Small Numbers in Scientific Notation**Question Bank 3.09 **ANSWER KEY**

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*Show your work. Circle your final answer.***For questions 1 through 4, write each number in scientific notation.**

1. 0.0005

$$5 \times 10^{-4}$$

2. 0.0000000049

$$4.9 \times 10^{-9}$$

3. 0.0316

$$3.16 \times 10^{-2}$$

4. 0.0000000000529 meters (the radius of a hydrogen atom)

$$5.29 \times 10^{-11} \text{ m}$$

*The 5 is in the eleventh decimal place.***For questions 5 and 6, write each number in standard form.**

- 5.
- $9 \times 10^{-6}$

$$0.000009$$

- 6.
- $1.8 \times 10^{-7}$

$$0.00000018$$

**Powers of ten.**

7. Write each value as a single power of 10, then as a fraction or decimal.

- a.
- $10^{-11}$
- , as a fraction

$$\frac{1}{100,000,000,000}$$

- b.
- $(10^4)^{-6} \times 10^{22}$

$$10^{-24} \cdot 10^{22} = 10^{-2} = 0.01$$

- c.
- $10^{-5} \cdot 10^8$

$$10^3 = 1,000$$

8. Two measurements are  $3.9 \times 10^{-5}$  and 0.000000074.

a. Write  $3.9 \times 10^{-5}$  in standard form.

**0.000039**

b. Write 0.000000074 in scientific notation.

**$7.4 \times 10^{-8}$**

c. Which of the two is greater? Explain how you know.

**$3.9 \times 10^{-5}$  — the exponent  $-5$  is greater than  $-8$ , so its power of 10 is larger;  $0.000039 > 0.000000074$ .**

9. Complete each comparison.

a. How many times larger is 9.6 than 3.2?

**3**

b. How many times larger is  $10^{-2}$  than  $10^{-6}$  ?

**$10^4 = 10,000$**

**$-2 - (-6) = 4$ .**

c. How many times larger is  $9.6 \times 10^{-2}$  than  $3.2 \times 10^{-6}$  ?

**$3 \times 10^4 = 30,000$  times larger**

10. Write each value in scientific notation.

a. 988 ten-thousandths

**$0.0988 = 9.88 \times 10^{-2}$**

b. 0.00988

**$9.88 \times 10^{-3}$**

c. 988 millionths

**$0.000988 = 9.88 \times 10^{-4}$**

11. Which is 0.00081 written in scientific notation?

☒ **A.  $8.1 \times 10^{-4}$**

☐ B.  $8.1 \times 10^{-3}$

☐ C.  $81 \times 10^{-5}$

☐ D.  $8.1 \times 10^4$

**A**

12. Select ALL of the expressions that are equal in value to 0.00052.

- ☒ A.  $5.2 \times 10^{-4}$
- ☒ B.  $52 \times 10^{-5}$
- ☒ C.  $0.52 \times 10^{-3}$
- ☒ D.  $520 \times 10^{-6}$
- ☐ E.  $5.2 \times 10^{-3}$
- ☐ F.  $5.2 \times 10^4$

**A, B, C and D**

*Only A is in scientific notation.*

13. Light travels about  $3 \times 10^8$  meters per second. A snail travels about  $3 \times 10^{-4}$  meters per second.

a. How many times faster is light? Write the answer as a power of 10 and in standard form.

$$10^{12} = 1,000,000,000,000 \text{ times}$$

$$3 \div 3 = 1 \text{ and } 8 - (-4) = 12.$$

b. How many times slower is the snail than light?

$$10^{12} \text{ times slower — the same comparison read the other way.}$$